ApontaUA

NON-FUNCTIONAL REQUIREMENTS

Collaborative Academic Document Management System

WORKING DRAFT

PROJECT TEAM

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Document Control

Project	ApontaUA — Collaborative Academic Document Management System
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1. Purpose and Scope

This document defines proposed non-functional requirements for ApontaUA. Unlike functional requirements, which describe what the system must do, non-functional requirements describe quality attributes, constraints and measurable expectations. This draft is intended to guide architecture, implementation and testing; priorities and targets should be reviewed by the project team and supervisors.

1.1 Priority Definitions

Must — required for the MVP. Should — important, but may be deferred if agreed by the team. Targets described as provisional must be validated through implementation and testing.

1.2 MVP Scale Assumption

The initial performance and load-testing scenario is a Department of Electronics, Telecommunications and Informatics (DETI), University of Aveiro, deployment with approximately 300 registered users. The team should define a representative number of concurrently active users and an initial document-volume assumption before final performance testing.

1.3 Availability Target

The MVP should be designed to remain usable and recover from common failures, but a formal uptime percentage is not fixed in this draft. The team should define and measure an availability target once the hosting environment, monitoring approach and maintenance assumptions are known.

2. Security and Access Control

ID	Requirement	Priority
NFR-01	The system shall enforce authorization on the server for every protected operation.	**Must**
NFR-02	If passwords are managed locally, they shall never be stored in plaintext.	**Must**
NFR-03	Communication between client and server shall use HTTPS in deployed environments.	**Must**
NFR-04	Uploaded files shall be checked for allowed type, size and basic safety before processing.	**Must**
NFR-05	Access to uploaded files and extracted document text shall be restricted according to permissions.	**Must**
NFR-06	Administrative and course-unit validation permissions shall follow the principle of least privilege.	**Must**

3. Privacy and Data Protection

ID	Requirement	Priority
NFR-07	Personal information shall be accessible only to users with appropriate permissions and a legitimate purpose.	**Must**

ID	Requirement	Priority
NFR-08	The project shall define retention and deletion rules for user data, uploaded files, extracted text, reports and audit records.	Must
NFR-09	Public material pages and search results shall not expose private user information unless explicitly authorized.	Must

4. Performance and Responsiveness

ID	Requirement	Priority
NFR-10	The team shall measure search and common-request response times under a documented load-test scenario; no strict two-second target is imposed at this stage.	Must
NFR-11	The system should provide visible status feedback when OCR, conversion or similarity analysis takes longer than an ordinary interactive request.	Must
NFR-12	OCR, file conversion and similarity detection should run without unnecessarily blocking unrelated user operations.	Must
NFR-13	Performance targets shall be revisited after the initial MVP implementation and measurements.	Must

5. Scalability and Capacity

ID	Requirement	Priority
NFR-14	The MVP shall be tested against a reference deployment of approximately 300 registered DETI users.	Must
NFR-15	The team shall define a representative number of concurrent users and an initial expected document volume for load testing.	Must
NFR-16	The system architecture should allow document-processing workloads to be scaled or moved to background workers if measurements show a need.	Should

6. Reliability, Recovery and Data Integrity

ID	Requirement	Priority
NFR-17	OCR, conversion or analysis failures shall not corrupt or replace the original uploaded file.	Must
NFR-18	The system shall preserve consistency between materials, metadata, version history and validation records.	Must
NFR-19	The project shall document a backup and recovery approach for stored materials and essential application data.	Must
NFR-20	Document-processing failures shall be logged sufficiently to support diagnosis by authorized personnel.	Must
NFR-21	The MVP shall be tested for recovery from common application and processing failures.	Should

7. Usability

ID	Requirement	Priority

ID	Requirement	Priority
NFR-22	The interface shall use consistent terminology and interaction patterns across the platform.	Must
NFR-23	The material-submission workflow shall clearly communicate required fields, errors and processing status.	Must
NFR-24	When OCR detects uncertain text, users shall be able to inspect the source image and correct the extracted text without re-entering the entire document.	Must
NFR-25	The interface shall clearly distinguish pending review, academically validated and not-yet-validated materials.	Must
NFR-26	Error messages shall explain the problem and, where possible, suggest a corrective action.	Must

8. Accessibility

ID	Requirement	Priority
NFR-27	The interface should follow WCAG 2.2 Level AA guidelines where applicable.	Should
NFR-28	Interactive functionality shall be operable using a keyboard.	Must
NFR-29	Meaningful images and non-text interface elements shall have appropriate text alternatives.	Must
NFR-30	Information shall not be conveyed by colour alone.	Must

9. Compatibility and File Handling

ID	Requirement	Priority
NFR-31	The MVP shall support uploads of PNG, JPG/JPEG, PDF, DOCX, PPTX, Markdown (.md) and XLSX files.	Must
NFR-32	The original uploaded file shall be preserved even when a preview or converted representation is generated.	Must
NFR-33	Upload/download support shall be distinguished from preview, text extraction and in-platform editing capabilities.	Must
NFR-34	Browser support shall be defined and tested during implementation based on the team's selected target browsers.	Must
NFR-35	Failure to generate a preview shall not prevent access to an otherwise accepted original file.	Must

10. Maintainability and Testability

ID	Requirement	Priority
NFR-36	The codebase shall be organized into modules with clearly defined responsibilities.	Must
NFR-37	Critical business logic, including permissions, validation workflows and duplicate detection, shall be covered by automated tests.	Must
NFR-38	The project shall use version control and a documented process for integrating code changes.	Must
NFR-39	Logs shall support diagnosis without unnecessarily exposing passwords, private document content or other sensitive information.	Must

ID	Requirement	Priority
NFR-40	Deployment and local development steps shall be documented and reproducible.	Should

11. OCR and Similarity Detection Quality

ID	Requirement	Priority
NFR-41	The OCR process shall identify uncertain text when confidence information is available and preserve the original document.	Must
NFR-42	Users shall be able to correct OCR errors before uncertain extracted text is treated as confirmed content.	Must
NFR-43	Exact-duplicate detection shall remain available even when text extraction fails.	Must
NFR-44	Similarity detection shall be evaluated on representative documents to estimate false-positive and false-negative rates.	Must
NFR-45	The system shall not treat a similarity score as definitive proof that two materials are duplicates.	Must
NFR-46	Moderation and OCR decisions shall be presented as automated assessments, not as guarantees of academic correctness.	Must

12. Auditability

ID	Requirement	Priority
NFR-47	The system shall keep traceable records of relevant administrative actions, moderation decisions and academic validation actions.	Must
NFR-48	Audit records shall identify the action, timestamp and responsible account where appropriate.	Must
NFR-49	Access to audit records shall be restricted to authorized roles.	Must

13. Availability and Operations

ID	Requirement	Priority
NFR-50	The MVP shall include a documented availability objective and a way to measure service interruptions before pilot use.	Must
NFR-51	The system should provide health checks or equivalent monitoring for critical application components.	Should
NFR-52	Planned maintenance and known service interruptions should be communicated to users where practical.	Could

14. Proposed Validation and Test Plan

- Load testing: use a DETI-like test dataset and a scenario representing approximately 300 registered users; separately choose and document concurrent-user assumptions.
- Performance: measure search, material metadata access, upload initiation and processing completion times. Record results before deciding on stricter service-level targets.
- Reliability: test recovery from OCR failure, conversion failure and interrupted processing while verifying that original files remain intact.

- Security: test server-side authorization, role boundaries, file validation and access restrictions.
- OCR and similarity quality: assemble representative typed PDFs, scanned pages, handwritten notes, iPad exports and office documents; measure recognition issues and duplicate-detection false positives/negatives.
- Availability: once hosting and monitoring are selected, define the measurement window, treatment of planned maintenance and target uptime.

15. Open Decisions

Topic	Decision to confirm
Availability target	Choose a measurable uptime target after hosting and monitoring are known.
Concurrent users	Define expected simultaneous active users within the 300 registered-user reference scenario.
Document volume	Estimate the number and average size of materials expected in the pilot.
Performance targets	Use early measurements to set realistic search and upload/processing objectives.
Browser support	Choose the supported browsers during implementation and test them explicitly.
Retention and deletion	Agree how long uploaded files, OCR text, reports and audit records are retained.
Accessibility validation	Choose how WCAG-related checks will be tested in the MVP.

16. Project Team and Supervision

Name	Role
Daniel Simbe	FullStack and DevOps Engineer
Anderson Tembe	FullStack Developer
Giovanni Santos	Backend and Architecture Engineer
Felizardo Samuel	QA Analyst and Frontend Developer
Marcelo Junior	Frontend Developer
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Appendix A — Interpretation

The 300-user figure is a reference for registered users, not a claim that all 300 users will be active concurrently. Concurrent load, document volume, availability percentage and strict response-time targets remain to be agreed and measured. This document intentionally avoids promising a performance level before the MVP has been tested.